

Topology of Deep Neural Networks

Overview

- ReLU fulfills the necessity for a non-injective, topologically invariant activation function
- Topological manifolds have uncountably many points in their sets, but in reality, we use a finite subset of the manifold and use persistent homology as a tool to give meaning to the finite subset and estimate the Betti numbers of the underlying manifold
- Footnote 2: “codes available for the reader’s further experimentation”
- Homeomorphic mappings preserve topology; nonhomeomorphic mappings, such as ReLU, can change topology
- In practice, homeomorphic mappings achieve topology changes through round-off errors
- We use $x \mapsto \max(Ax + b, 0)$, where $Ax + b$ moves any “topologically complicated” part of M outside the nonnegative orthant and ReLU collapses these into a single point

Quantifying Topology

- Betti numbers, $\beta = (\beta_0, \beta_1, \beta_2, \dots)$: count the number of k -dimensional holes of a manifold; loosely:
 - 0-dimensional holes: the number of connected components of a manifold
 - 1-dimensional holes: the number of independent loops that cannot be continuously shrunk to a point
 - 2-dimensional holes: the number of independent closed surfaces that enclose a 3-dimensional void
- Topological complexity: $\omega(M) = \beta_0(M) + \beta_1(M) + \beta_2(M) + \dots$
- Our goal is to track the Betti numbers

$$\beta(M) \rightarrow \beta(\nu_1(M)) \rightarrow \beta(\nu_2(M)) \rightarrow \dots \rightarrow \beta(\nu_{l-1}(M)) \rightarrow \beta(\nu_l(M)).$$

To do this in reality, we will have to estimate $\beta(M)$ from a point cloud, $X \subset M$, sampled from M , possibly with noise

Algebraic Topology and Persistent Homology Background

Simplicial Complexes

- Simplex, σ : (k -dimensional) the convex hull of $k + 1$ affinely independent points $v_0, \dots, v_k \in \mathbb{R}^d$
 - 0-dimensional: a point
 - 1-dimensional: a line segment
 - 2-dimensional: a triangle
 - 3-dimensional: a tetrahedron
- Simplicial Complex, K : (m -dimensional) a set of simplices of at most dimension m , satisfying the following:
 - (i) $\sigma_0 \subset \sigma \in K \implies \sigma_0 \in K$
 - (ii) $\sigma_1, \sigma_2 \in K, \sigma := \sigma_1 \cap \sigma_2 \neq \emptyset \implies \sigma \subset \sigma_1$ and $\sigma \subset \sigma_2$

Homology and Betti Numbers

Let $M \subset \mathbb{R}^d$ be a manifold and $k \in \{0, 1, \dots, d\}$:

- The k -th homology group of M is given by $H_k = \ker(\partial_k)/\text{im}(\partial_{k+1})$
- Homology classes are elements of H_k which form cosets/equivalence classes
- Then, $\dim(H_k) = \beta_k$, the k -th Betti number of M

Simplicial Homology

- Given an abstract simplicial complex, we define an $\mathbb{F}_2$ -vector space $C_k(K)$ as follows: Let $K^{(k)} = \{\sigma_1, \sigma_2, \dots, \sigma_m\}$ be the set of all k -dimensional simplices in K . Then, an element of $C_k(K)$ is a formal linear combination:

$$\sum_{j=1}^m n_j \sigma_j, \quad n_j \in \mathbb{F}_2.$$

Then, $C_k(K)$ is a vector space over $\mathbb{F}_2$ with $K^{(k)}$ as a basis.

- Boundary operator:

$$\begin{aligned} \partial_k : C_k(K) &\rightarrow C_{k-1}(K) \\ \sigma = [v_0, \dots, v_k] &\mapsto \sum_{j=1}^k [v_0, \dots, \hat{v}_j, \dots, v_k] \end{aligned}$$

where $\hat{v}_j$ indicates that v_j is omitted.

- Then, $\partial_k \left(\sum_{j=1}^m n_j \sigma_j \right) := \sum_{j=1}^m n_j \partial_k \sigma_j$, i.e. ∂_k is a linear operator.
- $\partial_1[a, b] = [b] + [a]$, $\partial_2[a, b, c] = [b, c] + [a, c] + [a, b]$, $\partial_2([a, b, c] + [d, e, f]) = \partial_2[a, b, c] + \partial_2[d, e, f]$
- $\beta_k(K) = \dim(H_k(K)) = \text{nullity}(\partial_k) - \text{rank}(\partial_{k+1})$, $k \in \{0, \dots, d\}$

Vietoris-Rips Complex

Let $X \subset M \subset \mathbb{R}$ be a point-cloud data set. Then,

$$\text{VR}_\varepsilon(X) := \{[x_0, \dots, x_k] : \delta(x_i, x_j) \leq 2\varepsilon, x_0, \dots, x_k \in X, k = 0, 1, \dots, n\},$$

i.e. the set of k -simplices such that to vertices are contained in a non-zero simplex if and only if they are within the prescribed distance to each other.

We need to choose ε and δ to create the simplicial-complex with the most information possible, without overfitting the dataset to a discrete set ($\varepsilon = 0$).

- δ is chosen as the geodesic distance in M , estimated from X using the graph geodesic distance
- to obtain ε_* , we use *persistent barcodes*

Persistent Homology

- Generally speaking, a persistence barcode is an interval $[\varepsilon, \varepsilon']$ where its left-end point ε is the scale at which the new feature appears or born, and its right-end point ε' is the scale at which that feature disappears or die. The length of the interval $\varepsilon' - \varepsilon$ is the persistence of that feature.
- Features that are non-robust to perturbations will produce short intervals; conversely, features that persist long enough, i.e., produce long intervals, are thought to be prominent features of the underlying manifold
- We are interested in selecting ε_* from a finite set of $\varepsilon_0 < \varepsilon_1 < \dots < \varepsilon_m$. Letting $K_j = \text{VR}_{\varepsilon_j}(X)$, we then have that $K_0 \subset K_1 \subset \dots \subset K_m$.

$$\begin{array}{cccccccccccccccccccc}
0 & \xrightarrow{\partial_{d+1}} & C_d(K_1) & \xrightarrow{\partial_d} & C_{d-1}(K_1) & \xrightarrow{\partial_{d-1}} & \dots & \xrightarrow{\partial_{k+1}} & C_k(K_1) & \xrightarrow{\partial_k} & C_{k-1}(K_1) & \xrightarrow{\partial_{k+1}} & \dots & \xrightarrow{\partial_2} & C_1(K_1) & \xrightarrow{\partial_1} & C_0(K_1) & \xrightarrow{\partial_0} & 0 \\
& & \downarrow f_1 & & \downarrow f_1 & & & & \downarrow f_1 & & \downarrow f_1 & & & & \downarrow f_1 & & \downarrow f_1 & & \\
0 & \xrightarrow{\partial_{d+1}} & C_d(K_2) & \xrightarrow{\partial_d} & C_{d-1}(K_2) & \xrightarrow{\partial_{d-1}} & \dots & \xrightarrow{\partial_{k+1}} & C_k(K_2) & \xrightarrow{\partial_k} & C_{k-1}(K_2) & \xrightarrow{\partial_{k+1}} & \dots & \xrightarrow{\partial_2} & C_1(K_2) & \xrightarrow{\partial_1} & C_0(K_2) & \xrightarrow{\partial_0} & 0 \\
& & \downarrow f_2 & & \downarrow f_2 & & & & \downarrow f_2 & & \downarrow f_2 & & & & \downarrow f_2 & & \downarrow f_2 & & \\
0 & \xrightarrow{\partial_{d+1}} & C_d(K_3) & \xrightarrow{\partial_d} & C_{d-1}(K_3) & \xrightarrow{\partial_{d-1}} & \dots & \xrightarrow{\partial_{k+1}} & C_k(K_3) & \xrightarrow{\partial_k} & C_{k-1}(K_3) & \xrightarrow{\partial_{k+1}} & \dots & \xrightarrow{\partial_2} & C_1(K_3) & \xrightarrow{\partial_1} & C_0(K_3) & \xrightarrow{\partial_0} & 0 \\
& & \downarrow f_3 & & \downarrow f_3 & & & & \downarrow f_3 & & \downarrow f_3 & & & & \downarrow f_3 & & \downarrow f_3 & & \\
\vdots & & \vdots & & \vdots & & \vdots & & \vdots & & \vdots & & \vdots & & \vdots & & \vdots & & \vdots \\
& & \downarrow f_{m-1} & & \downarrow f_{m-1} & & & & \downarrow f_{m-1} & & \downarrow f_{m-1} & & & & \downarrow f_{m-1} & & \downarrow f_{m-1} & & \\
0 & \xrightarrow{\partial_{d+1}} & C_d(K_m) & \xrightarrow{\partial_d} & C_{d-1}(K_m) & \xrightarrow{\partial_{d-1}} & \dots & \xrightarrow{\partial_{k+1}} & C_k(K_m) & \xrightarrow{\partial_k} & C_{k-1}(K_m) & \xrightarrow{\partial_{k+1}} & \dots & \xrightarrow{\partial_2} & C_1(K_m) & \xrightarrow{\partial_1} & C_0(K_m) & \xrightarrow{\partial_0} & 0
\end{array}$$

Overview of Problem and Methodology

- focus on binary classification
- “we seek to classify two different probability distributions supported on two disjoint manifolds $M_a, M_b \subset \mathbb{R}^d$ ”
- take $T \subset M_a \cup M_b = M$ uniformly and densely
- train the NN $\nu : \mathbb{R}^d \rightarrow [0, 1]$ on T until well-trained
- deliberately choose M_a and M_b to have topologically complicated decision boundary $D \subset \mathbb{R}^d$
- results will show that a well-trained NN reduces the topology on a layer-by-layer basis until D is a hyperplane in $\mathbb{R}^p$ in the last layer
- In our approach, we can only indirectly observe $\nu_j(D)$ using $\nu_j(M_a), \nu_k(M_b)$, but not directly, i.e. no explicit $\omega(D)$

Real Versus Simulated Data

- In short, we use both real and simulated data
- Simulated Data:
 - Has well-defined topological features that are known before-hand
 - Clean samples, so we can skip denoising (step (i))
 - Samples uniformly distributed on M
 - Can always simulate data with a perfect classifier
- Real-World Data:
 - can have vastly more complicated topologies that are nearly impossible to determine
 - may not have complete separation of categories, making it impossible to detangle
- Simulated data allow us to pick the right ε_* so we only need to compute homology at a single scale, whereas with real data we need to compute homology for a range of scales, making real data more computationally expensive
- In short, simulated data is easy, readily accessible, and non-trivial enough for this paper
- Hence, we do most of our experiments on simulated data and use some real world data at the end to validate the results from the simulation

Methodology

- 3 simulated data sets
- many, many NNs trained to the prescribed conditions
- Steps for computing k_* and ε_* :
 - (i) Let $\varepsilon = 1$
 - (ii) Choose k_* so that $\beta_0(\text{VR}_{k_*,1}(X)) = \beta_0(M)$
 - (iii) Let $k = k_*$
 - (iv) Choose ε_* so that $\beta_k(\text{VR}_{k_*,1}(X)) = \beta_k(M)$ for all $k \in \{1, 2, \dots, d-1\}$
- Key take away: Homology computations are way more expensive than training to near zero generalization error and can take up to 80 minutes for a single pair (k_*, ε_*)

Results and Discussions

- Topology changes and is simplified layer-by-layer
- Non-homeomorphic mappings (ReLU) reduce topology faster than homeomorphic mappings (tanh)
- tanh struggles to change topology/requires more effort
- Network width does not generally have a significant effect on topological changes, but very narrow networks are much harder to train to a high accuracy
- Reducing network depth forces topological changes to concentrate in final layers rather than being evenly distributed. Network must be of sufficient length to change topology based on complexity of original data set

Consistency with Real-World Data

- Results from simulated experiments tested on 4 real-world data sets
- All experiments confirmed the same results that neural networks reduce topological complexities in our data and that non-homeomorphic mappings reduce topology faster than homeomorphic mappings

Conclusion

- Restatement of results
- We concluded that ReLU changes topology faster than homeomorphic mappings, but did not investigate the explicit mechanisms of how ReLU changes topology. We speculate that while affine mappings allow for translations, reflections, and contractions, composition with ReLU add the ability to fold space

Planning

- Visualization: write a Python script to visualize a data set over an underlying manifold and allow the user to select an ϵ value for the Vietoris-Rips complex. Should have a 3d visual.